

MICROSTRIP TRANSMISSION LINES

file: mstrip.mcd

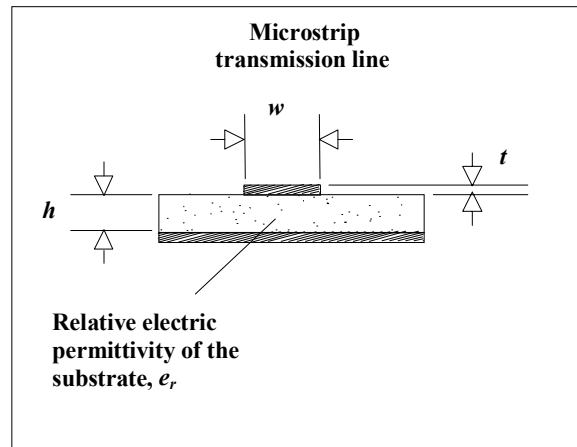
Formulas included in this spreadsheet:

Effective relative permittivity	EEFF() (used internally)
Effective electrical trace width	WE() (used internally)
Microstrip characteristic impedance	ZMSTRIP()
Microstrip propagation delay	PMSTRIP()
Microstrip trace inductance	LMSTRIP()
Microstrip trace capacitance	CMSTRIP()

Formulas from: I. J. Bahl and Ramesh Garg, "Simple and accurate formulas for microstrip with finite strip thickness", Proc. IEEE, 65, 1977, pp. 1611-1612.

This material is nicely summarized in T. C. Edwards, "Foundations of Microstrip Circuit Design," John Wiley, New York, 1981, reprinted 1987.

(Watch out for Edward's error in Equation 3.52b, where he omits a $\ln()$ function.)



mstrip

Variables used:

h	Trace height above ground (in.)
w	Trace width (in.)
t	Trace thickness (in.)
e_r	Relative permittivity of material between trace and ground plane (dimensionless)
x	Trace length (in.)

Effective relative permittivity
as a function of microstrip trace geometry:

For skinny traces ($w < h$)

$$E_{\text{skny}}(h, w, \epsilon_r) := \frac{\epsilon_r + 1}{2} + \left(\frac{\epsilon_r - 1}{2} \right) \cdot \left[\left(1 + \frac{12 \cdot h}{w} \right)^{-.500} + .04 \cdot \left(1 - \frac{w}{h} \right)^2 \right]$$

For wide traces ($w > h$)

$$E_{\text{wide}}(h, w, \epsilon_r) := \frac{\epsilon_r + 1}{2} + \left(\frac{\epsilon_r - 1}{2} \right) \cdot \left(1 + \frac{12 \cdot h}{w} \right)^{-.500}$$

Composite formula picks skinny or wide model depending on w/h ratio:

$$E_{\text{temp}}(h, w, \epsilon_r) := \text{if}(w > h, E_{\text{wide}}(h, w, \epsilon_r), E_{\text{skny}}(h, w, \epsilon_r))$$

Special adjustment to account for trace thickness:

$$EEFF(h, w, t, \epsilon_r) := E_{\text{temp}}(h, w, \epsilon_r) - \frac{(\epsilon_r - 1) \cdot \left(\frac{t}{h} \right)}{4.6 \cdot \sqrt{\frac{w}{h}}}$$

When w/h is skinny, you get the average of the
PCB permittivity, ϵ_r , and the permittivity of air.
When w/h is wide, (the trace is very close to the
ground plane) you get ϵ_r .

Effective trace width as a function
of other parameters (in.):

For skinny traces ($2\pi w < h$)

$$WE_skny(h, w, t) := w + \frac{1.25 \cdot t}{\pi} \cdot \left(1 + \ln \left(\frac{4 \cdot \pi \cdot w}{t} \right) \right)$$

For wide traces ($2\pi w > h$)

$$WE_wide(h, w, t) := w + \frac{1.25 \cdot t}{\pi} \cdot \left(1 + \ln \left(\frac{2 \cdot h}{t} \right) \right)$$

Composite formula picks skinny or wide model depending on w/h ratio:

$$WE(h, w, t) := \text{if} \left(w > \frac{h}{2 \cdot \pi}, WE_wide(h, w, t), WE_skny(h, w, t) \right)$$

Characteristic impedance as a function
of trace geometry (Ω):

Accuracy of better than 2 percent is
obtained under the following conditions:

$$\begin{array}{l} 0 < t/h < 0.2 \\ 0.1 < w/h < 20 \\ 0 < \epsilon_r < 16 \end{array}$$

For skinny traces ($w < h$)

$$ZMS_skny(h, w, t) := 60 \cdot \ln \left(\frac{8 \cdot h}{WE(h, w, t)} + \frac{WE(h, w, t)}{4 \cdot h} \right)$$

For wide traces ($w > h$)

$$ZMS_wide(h, w, t) := \frac{120 \cdot \pi}{\frac{WE(h, w, t)}{h} + 1.393 + .667 \cdot \ln \left(\frac{WE(h, w, t)}{h} + 1.444 \right)}$$

Composite formula picks skinny or wide model depending on w/h ratio:

$$ZMSTRIP(h, w, t, \epsilon_r) := \frac{\text{if}(w > h, ZMS_wide(h, w, t), ZMS_skny(h, w, t))}{\sqrt{EEFF(h, w, t, \epsilon_r)}}$$

Microstrip propagation delay (s/in.):

$$\text{PMSTRIP}(h, w, t, \text{er}) := 84.72 \cdot 10^{-12} \cdot \sqrt{\text{EEFF}(h, w, t, \text{er})}$$

Inductance of microstrip (H):

$$\text{LMSTRIP}(h, w, t, x) := \text{PMSTRIP}(h, w, t, 1.) \cdot \text{ZMSTRIP}(h, w, t, 1.) \cdot x$$

(Use a dummy er value of 1. It doesn't matter for inductance calculations.)

Capacitance of microstrip (F):

$$\text{CMSTRIP}(h, w, t, \text{er}, x) := \frac{\text{PMSTRIP}(h, w, t, \text{er})}{\text{ZMSTRIP}(h, w, t, \text{er})} \cdot x$$

Example microstrip wire calculations

Height above ground (in.)	H := .006	
Width of trace (in.)	W := .008	
Thickness of trace (in.)	T := .00137	(1-oz copper plating weight)
Length of wire (in.)	X := 11.000	
Relative electric permeability (affects capacitance, but not inductance)	er := 4.5	

Impedance (Ω):

$$\text{ZMSTRIP}(H, W, T, \text{er}) = 56.4435$$

Total inductance (H):

$$\text{LMSTRIP}(H, W, T, X) = 9.3401 \times 10^{-8}$$

Same result in nH:

$$\text{LMSTRIP}(H, W, T, X) \cdot 10^9 = 93.4008$$

Inductance per in. (H):

$$\text{LMSTRIP}(H, W, T, 1) = 8.491 \times 10^{-9}$$

Total capacitance (F):

$$\text{CMSTRIP}(H, W, T, \text{er}, X) = 2.9317 \times 10^{-11}$$

Same result in pF:

$$\text{CMSTRIP}(H, W, T, \text{er}, X) \cdot 10^{12} = 29.3172$$

Capacitance per in. (F):

$$\text{CMSTRIP}(H, W, T, \text{er}, 1) = 2.6652 \times 10^{-12}$$

Tolerance effects

$$\text{ZMSTRIP_TOL}(h, dh, w, dw, t, er, der) := \begin{pmatrix} \text{ZMSTRIP}(h + dh, w - dw, t, er - der) \\ \text{ZMSTRIP}(h, w, t, er) \\ \text{ZMSTRIP}(h - dh, w + dw, t, er + der) \end{pmatrix}$$

$$\alpha := \text{ZMSTRIP_TOL}(.007, .002, .011, .002, .0022, 4.5, .1)$$

$$\alpha = \begin{pmatrix} 64.7868 \\ 51.3724 \\ 37.9267 \end{pmatrix}$$

$$\text{REFL}(x, z) := \begin{pmatrix} \frac{z - x_0}{z + x_0} \\ \frac{z - x_1}{z + x_1} \\ \frac{z - x_2}{z + x_2} \end{pmatrix}$$

$$\text{REFL}(\alpha, 50) = \begin{pmatrix} -0.1288 \\ -0.0135 \\ 0.1373 \end{pmatrix}$$